

User Manual

Customer Segmentation Dashboard — For Business Users

1. Starting the Application

Ask your technical team to run the application (streamlit run app/app.py), or open the shared deployment link if one has been provided. The dashboard opens in your web browser automatically.

2. Navigating the Dashboard

Use the numbered sections in the left sidebar ("1. Executive Overview" through "10. Marketing Recommendations") to move between views. Each section focuses on a specific business question.

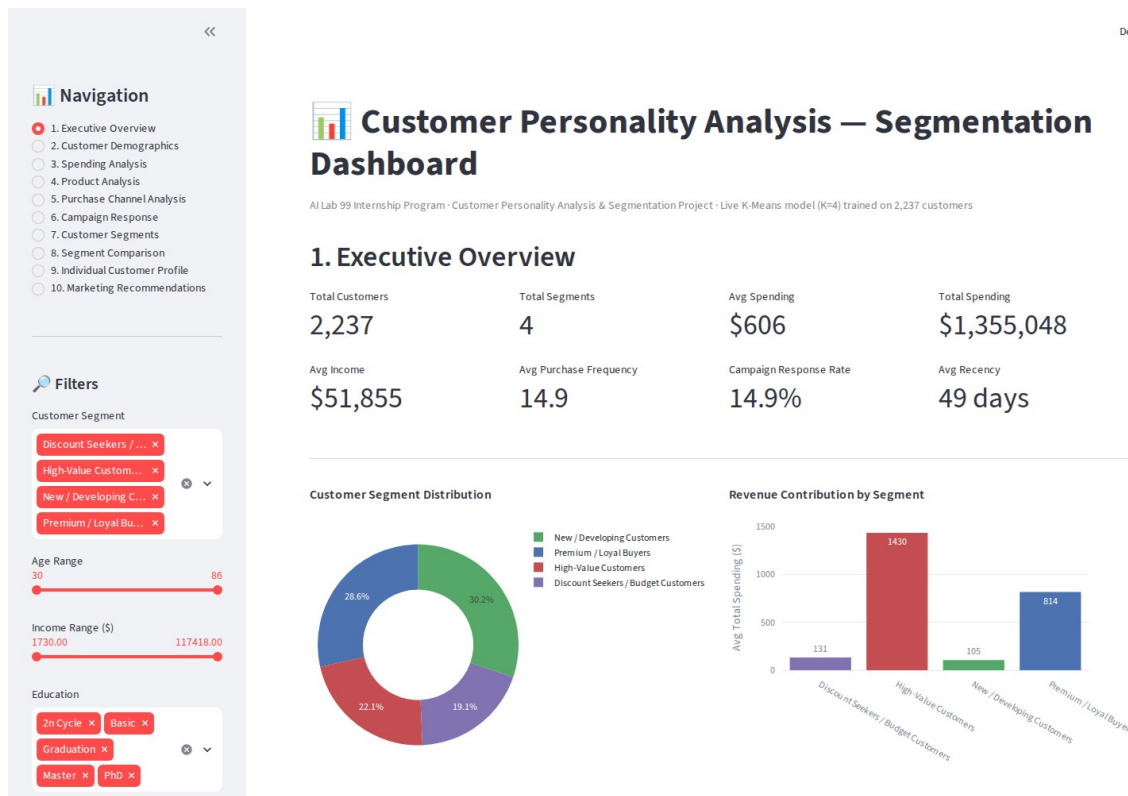


Figure 1 — The Executive Overview is the landing section, showing headline KPIs.

3. Applying Filters

The Filters panel on the left lets you narrow the customer base by segment, age range, income range, education, marital status, product category, purchase channel, and campaign response. Every chart and table on the page updates immediately when you change a filter — no need to click a separate 'apply' button.

4. Comparing Segments

Go to section 8 ("Segment Comparison"), choose Segment A and Segment B from the two dropdowns, and the radar chart plus the side-by-side metrics table update automatically. Use the download button to save the comparison as a CSV file.

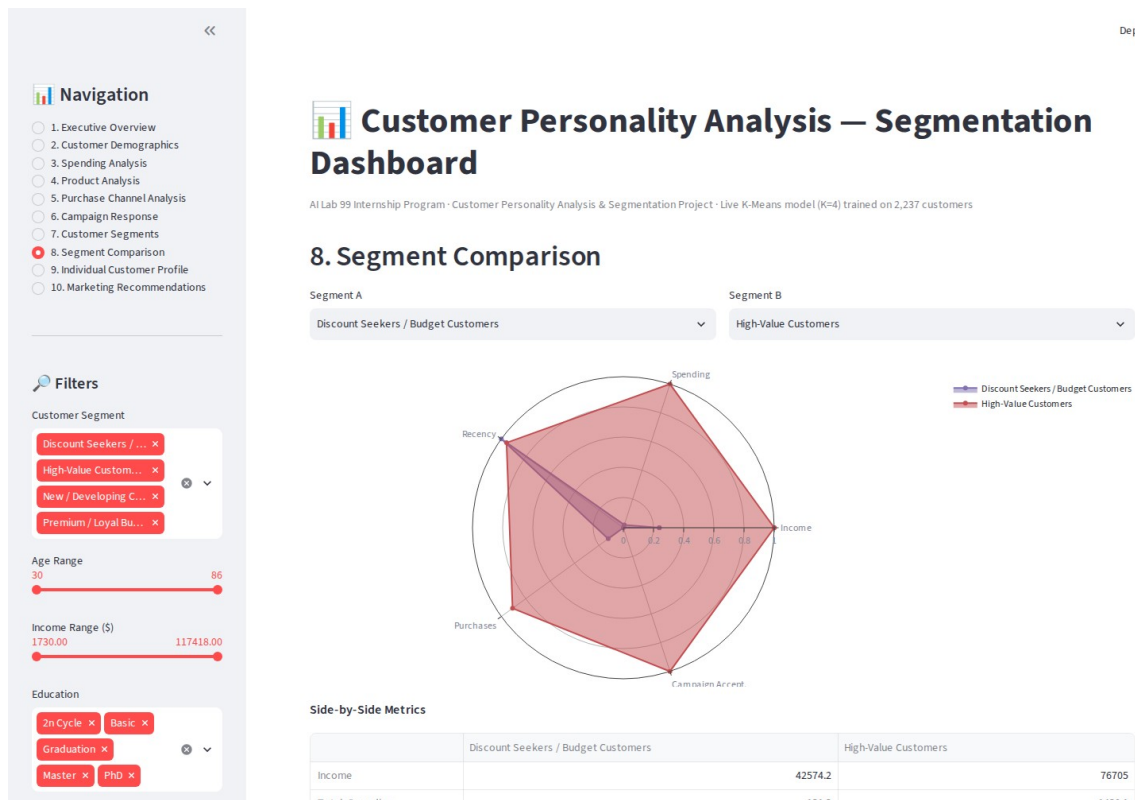


Figure 2 — Segment Comparison section with radar chart.

5. Searching for a Customer

Go to section 9 ("Individual Customer Profile"), open the "Look Up Existing Customer" tab, type a Customer ID, and click "Look Up Customer". Their segment, income, spending, and recommended marketing action appear immediately.

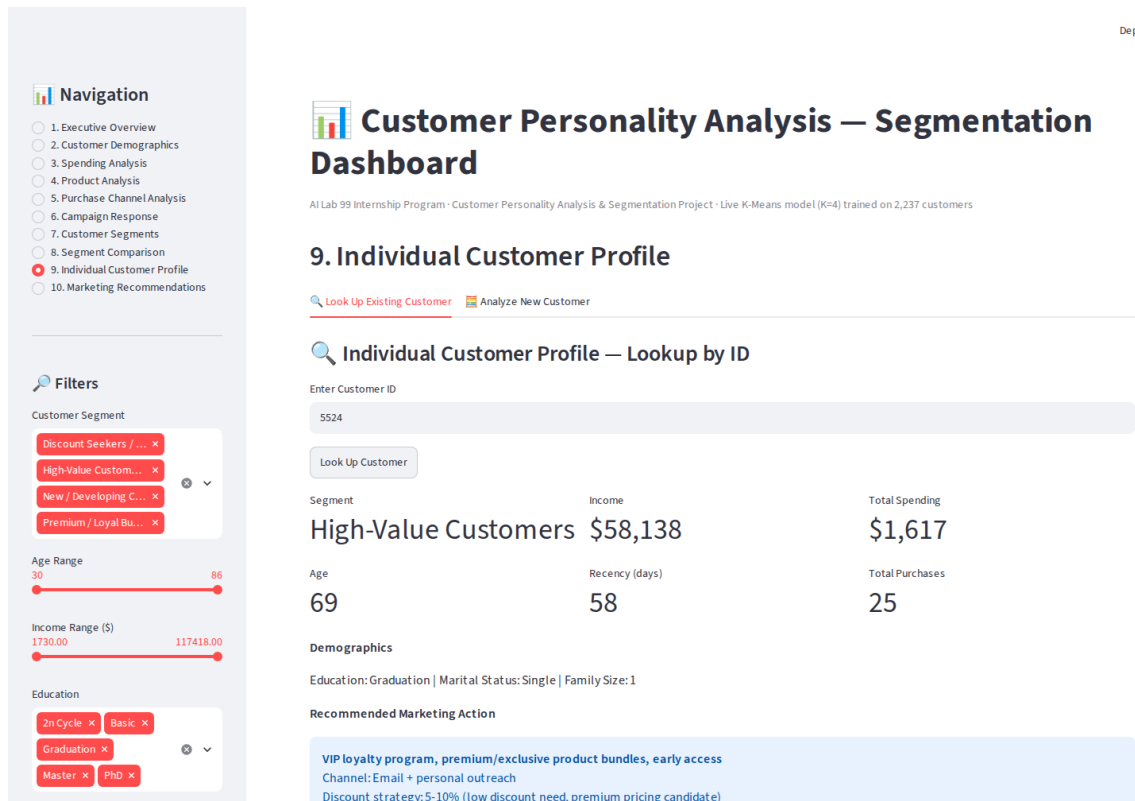


Figure 3 — Individual customer lookup by ID.

6. Analyzing a New Customer

In the same section, switch to the "Analyze New Customer" tab, fill in Age, Income, Total Spending, Web Purchases, Store Purchases, and Recency, then click "Analyze Customer". The dashboard runs the live trained model and returns the assigned segment, customer value, retention risk, and recommended action.

Navigation

- 1. Executive Overview
- 2. Customer Demographics
- 3. Spending Analysis
- 4. Product Analysis
- 5. Purchase Channel Analysis
- 6. Campaign Response
- 7. Customer Segments
- 8. Segment Comparison
- 9. Individual Customer Profile
- 10. Marketing Recommendations

Filters

Customer Segment

- Discount Seekers / ...
- High-Value Custom...
- New / Developing C...
- Premium / Loyal Bu...

Age Range

30 86

Income Range (\$)

1730.00 117418.00

Education

- 2n Cycle
- Basic
- Graduation
- Master
- PhD

Customer Personality Analysis — Segmentation Dashboard

AI Lab 99 Internship Program · Customer Personality Analysis & Segmentation Project · Live K-Means model (K=4) trained on 2,237 customers

9. Individual Customer Profile

Look Up Existing Customer **Analyze New Customer**

Analyze a New Customer

Enter customer details below. The trained K-Means model will assign a segment live.

Age	Total Spending (\$)	Store Purchases
35	1200	8
Income (\$)	Web Purchases	Recency (days)
50000	5	20

Analyze Customer

Customer analyzed successfully.

Customer Segment	Customer Value	Retention Risk
New / Developing C...	Low (growth potent...	High

Recommended Action

Welcome discount, starter bundle, onboarding email/SMS series

Figure 4 — Live segment prediction for a new customer.

7. Interpreting Customer Segments

Each segment has a consistent color throughout the dashboard: red for High-Value Customers, blue for Premium/Loyal Buyers, purple for Discount Seekers/Budget Customers, and green for New/Developing Customers. Section 7 ("Customer Segments") lets you expand each segment to see its defining characteristics.

8. Viewing Marketing Recommendations

Section 10 ("Marketing Recommendations") lists the recommended action, channel, and discount strategy for every segment, generated from the Module 08 business analysis.

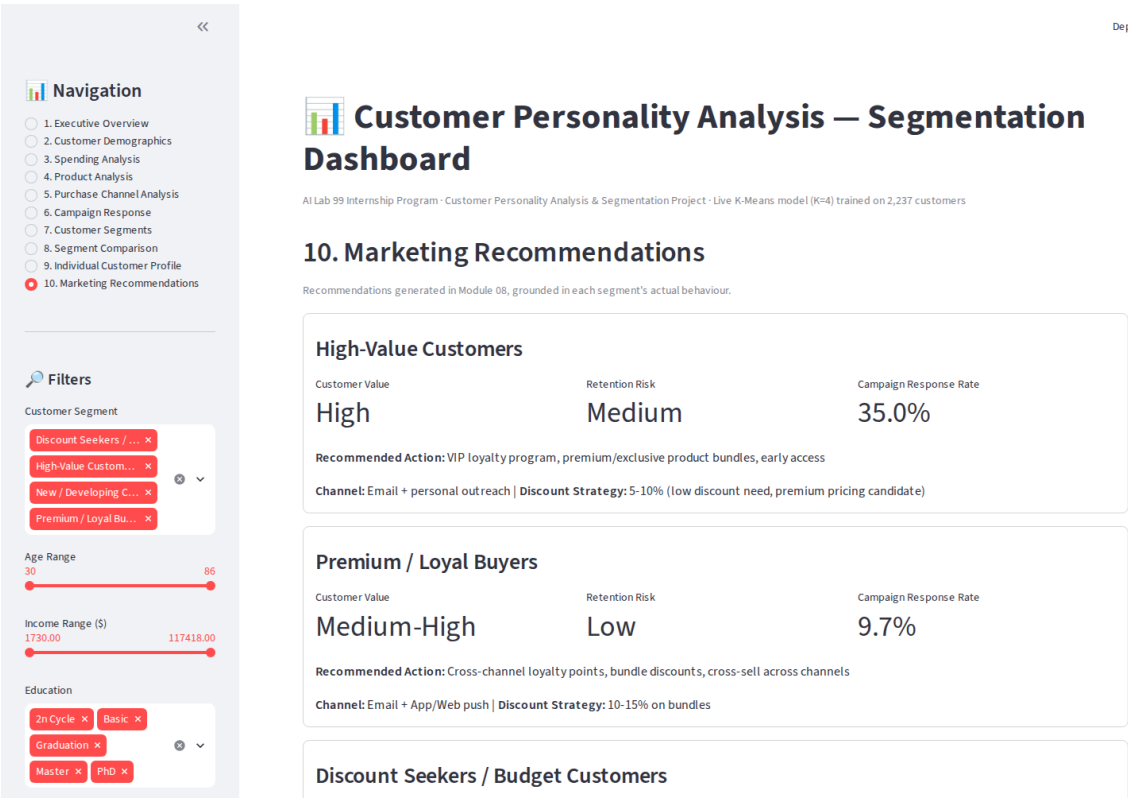


Figure 5 — Marketing Recommendations section.

9. Interpreting Charts

Every chart title states the business question it answers (e.g. "Which segment spends the most, on average?"). Hover over any chart element to see exact values. Box plots show the median (center line), interquartile range (box), and outliers (dots).